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Bookstore Management System

Software Requirement Specification

Submission Profile

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1. Introduction

1.1. Purpose

This document lists what the Bookstore Management System (BMS) must do and how well it must do it. It is the main guide for our project. Every requirement is linked to at least one planned test case, and the important requirements have acceptance criteria (pass rules), so we can prove the finished software is correct.

1.2. Scope

BMS is a desktop program for one shop: Qadri Book Store, Shahi Bazaar, Rohri. It replaces notebooks and paper bills. It will:

- Keep products and stock (add, change, delete, search, low-stock alert)
- Make cash sales with bills and receipts
- Handle school bulk orders and quotations
- Manage suppliers, purchase orders and goods received
- Give sales and stock reports, user accounts, backup and restore

Not included: online shopping, online payments (all sales are cash), many branches, AI forecasting and mobile apps.

1.3. Product Overview

BMS is a stand-alone Windows desktop application. It works fully offline and keeps all data on the shop's own computer. It has seven parts: Login and Roles, Inventory, Sales (POS), School Orders, Suppliers, Reports and System Admin.

1.4. Users

User	Who	What they do
Admin	Store owner / manager	Uses every feature of BMS.
Staff	Counter worker	Makes sales, prints receipts, searches products.
School	Outside, bulk buyer	Gets quotations. Does not use BMS.
Supplier	Outside, gives books	Gets purchase orders. Does not use BMS.
Walk-in Customer	Outside, buys at the counter	Gets a receipt. Does not use BMS.

1.5. Working Environment

- Runs on the shop's counter PC (Windows 10 or 11) with keyboard, mouse and a thermal receipt printer. The PC is low-spec, so BMS must be light.
- Needs no internet, because the shop's internet is not reliable.
- Built with Java 25 and Spring Boot (logic), Next.js with TypeScript (screens), Electron (desktop app) and SQLite (local database).

1.6. Rules and Assumptions

- All sales are cash. Every product has a unique SKU and a minimum stock level (used for low-stock alerts).
- School and supplier orders must be saved before they are processed.
- Users know basic computer use and type data correctly.

2. Functional Requirements

Each requirement starts with “The system shall …”.

ID	The system shall …	Priority
Login and Roles		
FR-01	allow users to log in with a username and password.	High
FR-02	limit features by role (Admin or Staff).	High
FR-03	allow users to log out.	Medium
Inventory		
FR-04	allow Admin to add a new product.	High
FR-05	allow Admin to change product details.	Medium
FR-06	allow Admin to delete a product.	Medium
FR-07	allow users to search and view products.	High
FR-08	show the current stock of each product.	Medium
FR-09	show an alert for low-stock products.	High
Sales (POS)		
FR-10	allow users to make a sale.	High
FR-11	allow users to add items to a sale.	High
FR-12	calculate the total cost automatically.	High
FR-13	update stock after each sale.	High
FR-14	create a receipt for each sale.	High
School Orders		
FR-15	allow Admin to register a school.	Medium
FR-16	allow Admin to create a school order.	Medium
FR-17	create a quotation for a school order.	Medium
FR-18	allow tracking of order status.	Medium
Suppliers		
FR-19	allow Admin to add a supplier.	Medium
FR-20	allow Admin to create a purchase order.	Medium
FR-21	record goods received from suppliers.	Medium
Reports		
FR-22	create sales reports.	Medium
FR-23	create inventory reports.	Medium
System Administration		
FR-24	allow Admin to manage user accounts.	Medium
FR-25	back up the data.	High
FR-26	restore data from a backup.	High
FR-27	print receipts and reports.	Medium

3. Non-Functional Requirements

These say how well the system must work. Each one can be measured.

ID	Requirement
Performance and Accuracy	
NFR-01	Screens, search and buttons respond within 3 seconds.
NFR-02	A sale finishes (confirm to receipt) within 5 seconds.
NFR-03	The system handles up to 200 sales a day without errors.
NFR-04	All totals, stock numbers and reports are correct.
Reliability	
NFR-05	The system asks for confirmation before anything is deleted.
NFR-06	Wrong or incomplete data is not saved.
NFR-07	Sales and orders are saved automatically.
NFR-08	Backup and restore do not lose any data.
NFR-09	The system works fully without internet.
Security	
NFR-10	Users must log in. Passwords are saved in hidden (hashed) form. Never as plain text
NFR-11	Each user can reach only the features of their role.
NFR-12	Important actions (login, sales, deletes) are logged with user and time.
Usability	
NFR-13	The screens are simple for users with basic computer skills.
NFR-14	Labels are clear. Every action shows a success, error or confirmation message, and error messages say what to do.

4. Use Cases

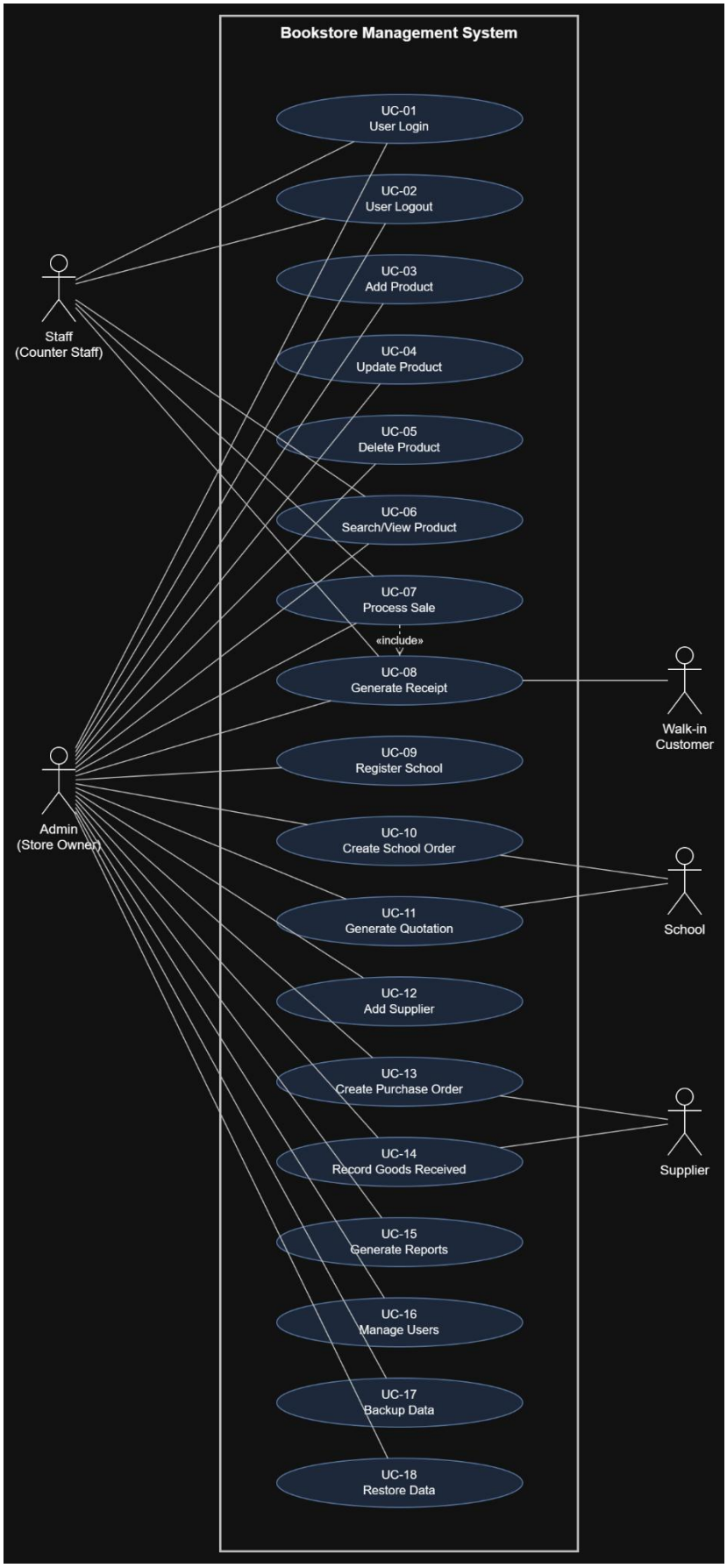
4.1. User Stories

ID	User Story	Requirements
US-01	As a user (Admin or Staff), I want to log in with my username and password, so that only allowed people use the system.	FR-01, FR-02
US-02	As a user, I want to log out, so that nobody else can use my account.	FR-03
US-03	As an Admin, I want to add a new product, so that I can sell it and track its stock.	FR-04
US-04	As an Admin, I want to change a product's details, so that its price and stock are always correct.	FR-05
US-05	As an Admin, I want to be asked "Are you sure?" before deleting a product, so that I do not delete it by mistake.	FR-06
US-06	As a user, I want to search for a product by name or SKU, so that I can find its price and stock quickly.	FR-07, FR-08
US-07	As an Admin, I want a low-stock alert, so that I can reorder before a book runs out.	FR-09
US-08	As a Staff member, I want to add items to a sale and see the total automatically, so that billing is fast and has no mistakes.	FR-10, FR-11, FR-12
US-09	As a Staff member, I want stock to update after each sale, so that stock numbers stay right.	FR-13
US-10	As a Staff member, I want to print a receipt for each sale, so that the customer has proof of purchase.	FR-14, FR-27
US-11	As an Admin, I want to register a school, so that I can take its bulk orders.	FR-15
US-12	As an Admin, I want to create a school order, so that bulk orders are recorded properly.	FR-16
US-13	As an Admin, I want to make a quotation for a school order, so that the school knows the total price before buying.	FR-17
US-14	As an Admin, I want to track the status of each school order, so that I know which orders are done.	FR-18
US-15	As an Admin, I want to add suppliers and create purchase orders, so that I can buy books and stationery in an organized way.	FR-19, FR-20
US-16	As an Admin, I want to record goods received, so that stock goes up when supplier items arrive.	FR-21
US-17	As an Admin, I want sales and inventory reports, so that I can see how the business is doing.	FR-22, FR-23, FR-27
US-18	As an Admin, I want to manage user accounts, so that only my workers can use the system.	FR-24
US-19	As an Admin, I want to back up and restore data, so that I do not lose records if the computer fails.	FR-25, FR-26

4.2. Use Case List

UC ID	Use Case	Actor(s)	Related FRs
UC-01	User Login	Admin, Staff	FR-01, FR-02
UC-02	User Logout	Admin, Staff	FR-03
UC-03	Add Product	Admin	FR-04
UC-04	Update Product	Admin	FR-05
UC-05	Delete Product	Admin	FR-06
UC-06	Search/View Product	Admin, Staff	FR-07, FR-08, FR-09
UC-07	Process Sale	Staff, Admin	FR-10, FR-11, FR-12, FR-13
UC-08	Generate Receipt	Staff, Admin	FR-14, FR-27
UC-09	Register School	Admin	FR-15
UC-10	Create School Order	Admin	FR-16, FR-18
UC-11	Generate Quotation	Admin	FR-17
UC-12	Add Supplier	Admin	FR-19
UC-13	Create Purchase Order	Admin	FR-20
UC-14	Record Goods Received	Admin	FR-21
UC-15	Generate Reports	Admin	FR-22, FR-23, FR-27
UC-16	Manage Users	Admin	FR-24
UC-17	Backup Data	Admin	FR-25
UC-18	Restore Data	Admin	FR-26

4.3. Use Case Diagram



5. Acceptance Criteria

These are the pass rules for the important (High priority) requirements and the key non-functional requirements. A requirement passes only when every rule is true.

ID	Acceptance Criteria
FR-01	1. A correct username and password opens the dashboard. 2. A wrong password, wrong username or empty field shows an error and stays on the login screen.
FR-02	1. When Staff logs in, the Users, Suppliers, Reports and Backup screens are not available. 2. When Admin logs in, all screens are available.
FR-04	1. A product with all required fields is saved and can be found by search. 2. A product with an empty required field is not saved and a message shows. 3. A product with a repeated SKU is not saved and a message shows.
FR-07	1. Searching by name or SKU shows the matching products with price and stock. 2. A search with no match shows “not found”.
FR-09	1. When quantity is equal to or below the minimum level, the low-stock alert shows. 2. When quantity is above the minimum level, no alert shows.
FR-10	1. A sale with at least one item can be confirmed and saved. 2. A sale with no items cannot be confirmed.
FR-11	1. An added item appears in the sale list with its quantity. 2. A quantity above the available stock shows a warning.
FR-12	The total equals the sum of (price × quantity) of all items. Example: 2 items at Rs. 100 and 1 item at Rs. 50 gives Rs. 250.
FR-13	After a sale of 3 units of a product that had 10 in stock, the stock shows 7.
FR-14	The receipt shows the items, quantities, total and date of the sale.
FR-25	A backup file is created and can be copied to a USB drive.
FR-26	After a restore, the data is the same as in the backup (same number of products and sales).
NFR-01	Every tested screen, search and button responds in 3 seconds or less.
NFR-02	The timed result from confirming a sale to the receipt is 5 seconds or less.
NFR-03	200 test sales are completed with 0 errors.
NFR-04	Every tested total, stock number and report matches the hand-calculated result.
NFR-10	No feature opens without login. The database shows no plain-text passwords.
NFR-11	Staff is blocked from Admin features in every tested case.

6. Requirements Traceability

This table links every requirement to the test cases that check it. TC means test case. Each test case is defined here by its title, and the full steps will be written when we prepare the tests. Every requirement has at least one test case.

Requirement	Test Case(s)
FR-01 Log in	TC-01 Valid login; TC-02 Wrong or empty details
FR-02 Roles	TC-03 Staff opens an Admin screen (blocked); TC-04 Admin opens all screens
FR-03 Log out	TC-05 Log out, then try to use a feature
FR-04 Add product	TC-06 Add product with full details; TC-07 Missing field or repeated SKU
FR-05 Change product	TC-08 Change product details
FR-06 Delete product	TC-09 Delete and confirm; TC-10 Delete and cancel
FR-07 Search product	TC-11 Search for an existing product; TC-12 Search for a missing product
FR-08 Show stock	TC-13 Compare stock shown with saved quantity
FR-09 Low-stock alert	TC-14 Quantity equal to minimum (alert); TC-15 Quantity above minimum (no alert)
FR-10 Make a sale	TC-16 Complete a sale with items; TC-17 Try a sale with no items
FR-11 Add items	TC-18 Add items to a sale; TC-19 Add a quantity above stock
FR-12 Total cost	TC-20 Check a sale total by hand; TC-21 Quantity 1 and maximum stock
FR-13 Update stock	TC-22 Check stock after a sale
FR-14 Receipt	TC-23 Check receipt contents
FR-15 Register school	TC-24 Register a school; TC-25 Register without a name
FR-16 School order	TC-26 Order for a registered school; TC-27 Order with an unavailable item
FR-17 Quotation	TC-28 Generate a quotation
FR-18 Order status	TC-29 Change and view order status
FR-19 Add supplier	TC-30 Add a supplier
FR-20 Purchase order	TC-31 Create a purchase order
FR-21 Goods received	TC-32 Record goods received (stock goes up)
FR-22 Sales report	TC-33 Sales report for chosen dates
FR-23 Inventory report	TC-34 Inventory report
FR-24 User accounts	TC-35 Add, change and remove a Staff account
FR-25 Backup	TC-36 Create a backup
FR-26 Restore	TC-37 Restore from a backup
FR-27 Printing	TC-38 Print a receipt and a report
NFR-01 3-second response	TC-39 Time screens, search and buttons

NFR-02 5-second sale	TC-40 Time a sale from confirm to receipt
NFR-03 200 sales a day	TC-41 Run 200 sales
NFR-04 Correct results	TC-20; TC-42 Check stock and report figures by hand
NFR-05 Delete confirmation	TC-09; TC-10
NFR-06 Wrong data rejected	TC-07; TC-43 Enter letters and negative numbers
NFR-07 Auto-save	TC-44 Close and reopen the app after a sale
NFR-08 No data loss	TC-36; TC-37
NFR-09 Works offline	TC-45 Use all features with the internet off
NFR-10 Login and passwords	TC-01; TC-46 Look for plain-text passwords in the database
NFR-11 Role access	TC-03; TC-47 Staff tries to reach an Admin feature another way
NFR-12 Logging	TC-48 Check the log after login, sale and delete
NFR-13 Simple screens	TC-49 A new user makes a sale with no help
NFR-14 Clear labels and messages	TC-02; TC-50 Check labels and messages on every screen

THE END